Constructing states from observations

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(Thanks to Dale, Jiamin, Bryan, Dikshant, and Mahvash!)

A simpler problem

Input/output data:

we know f^*

$X_t \in \mathcal{X}$

$$X_0, f^*(X_0)$$
$$X_1, f^*(X_0, X_1)$$
$$X_2, f^*(X_0, X_1, X_2)$$
$$\vdots$$

Goal: find $u(\cdot, \cdot)$ and $\hat{f}(\cdot)$

$$Z_{t+1} = u(X_t, Z_t)$$
$$\hat{f}(Z_{t+1}) \approx f^*(X_0, \ldots, X_{t+1})$$

Think about:

$\boxed{\text{size}(Z_t)}$ in terms of $\boxed{\text{size}(X_i)}$ and $\boxed{\text{size}(\mathcal{X})}$

Examples (constant space)



(1) Remember the first element

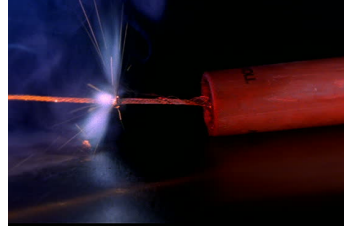
$$f^*(X_0, X_1, \ldots, X_{t-1}, X_t) = X_0 + X_t$$

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(2) Sliding window

$$f^*(X_0, \ldots, X_{t-3}, X_{t-2}, X_{t-1}, X_t) = X_{t-2} + X_{t-1} + X_t$$

Example (~constant space)



(3) Older elements have lower importance

$$f^*(X_0) = X_0$$
$$f^*(X_0, X_1) = X_1 + X_0^2$$
$$f^*(X_0, X_1, X_2) = X_2 + X_1^2 + X_0^3$$
$$f^*(X_0, X_1, X_2, X_3) = X_3 + X_2^2 + X_1^3 + X_0^4$$
$$\vdots$$

$$X_i \in \left[0, \frac{1}{e}\right]$$

$$f^*(X_0, \ldots, X_t) = X_t + \cdots + X_{t-w+1}^w + \underbrace{X_{t-w}^{w+1} + \cdots + X_0^{t+1}}_{\lesssim (1/e)^w}$$

$$w \gtrsim \log(1/\varepsilon) \quad \Rightarrow \quad \text{error} < \varepsilon$$

Examples (more than const. space)



(4) Need everything so far $(\Omega(t))$

$$f^*(X_0, \ldots, X_t) = (X_0, \ldots, X_t)$$

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(5) Median $(\Omega(t) \text{ or } \Omega(\log t))$

$$f^*(X_0, \ldots, X_t) = \mathsf{median}(X_0, \ldots, X_t)$$

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(6) Most frequent element $(\Omega(N))$

$$f^*(X_0, \ldots, X_t) = \mathsf{majority}(X_0, \ldots, X_t)$$

$$X_i \in \{1, 2, \ldots, N\}$$

Keyword: streaming and sketching algorithms.

Notice these contrasting views

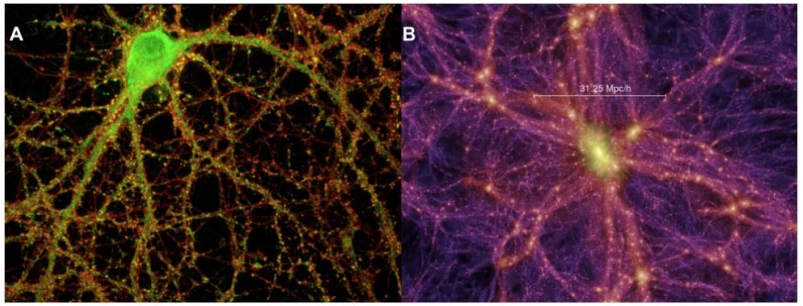
$$Z_t = (X_0, X_1, \ldots, X_t)$$

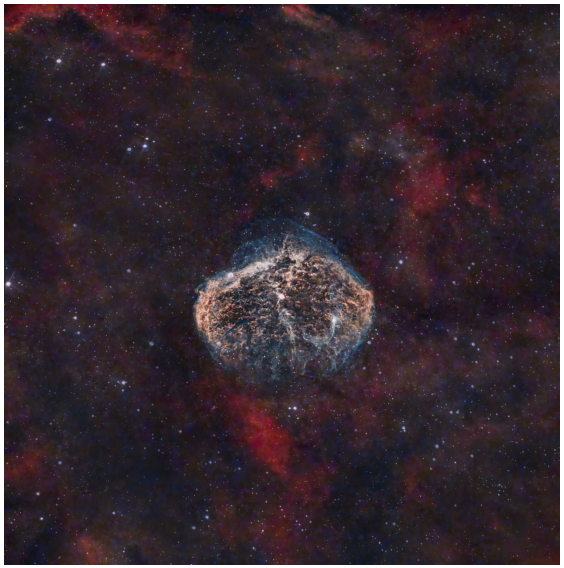
is not a good function for constructing (agent) states,
because the state grow unboundedly with time.

(Sutton & Barto, 2018; Sec. 17.3: Obvservation and State, RL book, 2/e)

Human brain has an "infinite" long-term memory







https://www.scientificamerican.com/article/what-is-the-memory-capacity/
https://www.universetoday.com/articles/one-of-these-pictures-is-the-brain-the-other-is-the-universe-can-you-tell-which-is-which
https://my.clevelandclinic.org/health/articles/long-term-memory

I would want to discuss with you all:

1. Why do we want fixed-sized states?

2. What environments support fixed-sized (agent) states? (amenable data distributions, simple target functions, etc.)

2. How to implement agents with growing state sizes? (embrace approximation, randomization, increasing compute/space req., etc.)

3. Additional issues in regression and RL.

Thank you!